

Supplementary Methods and Audit Archive for “From Predictive Importance to Decision Value: Evaluating XAI-Based Weights for ITS Prioritisation”

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1 Status and Interpretation Boundary

This supplement reports executed development audits and preserves the detailed primary and robustness analyses that had been specified before the prospectively frozen Pilot-B benchmark-adequacy gate was executed. The latter analyses were conditional on all five Pilot-B hard-stop flags being false. The modal-winner effective-tie stop fired, so the primary factorial was not authorized. Primary seeds 11001–11030 and reserve seeds 30001–30005 were never accessed; all sections explicitly labelled “Archived” remain unexecuted protocols rather than results.

The closed v1 state is anchored by an immutable version-control reference. Exact repository identifiers and complete digests are withheld from this anonymous reviewer file; they are available confidentially to editors and will accompany the public reproducibility archive after review. This document is therefore a methods, audit-results and archival-protocol supplement. It is not authorization to execute the protected design. After permanent Pilot-B closure, one separately frozen development-only descriptive diagnostic reconstructed the context-level MOORA selection maps of the seven observed archived fixed vectors. It used no primary/reserve world, generated no target outcome, fitted or queried no model, re-estimated no weight and could not alter the gate. Its protocol, implementation, independent audit and result were frozen as a distinct post-closure layer. Methodological citations are provided in the main manuscript and are not duplicated here.

Question Structure and Evidential Status

The main manuscript states two prospective research questions and one separate post-closure descriptive question. RQ1 asks whether structural diagnostics can reveal that apparent contextual variation disappears under the normalization used by the downstream MCDM operator. RQ2 asks whether, after that structural defect is corrected, structured global-weight pipelines provide sufficient incremental decision value over random-weight and constant-decision references to justify protected primary evaluation. Both are addressed by the prospectively structured benchmark-development record and the frozen Pilot-B result reported here.

DQ1 asks what the archived context-level selection maps reveal about the observed decision behaviour of the seven frozen fixed-weight vectors. DQ1 is post-closure and descriptive. It is non-confirmatory, it carries no pass/fail role, and it cannot modify, reinterpret or reopen the Pilot-B gate. Section 5.6 reports it under that label.

Throughout this supplement, *each replication seed defines one benchmark world*, and the world is the statistical unit. Where a numbered identifier such as 21001 appears it names a world; the phrase “replication seed” is retained only when the frozen artefact field or protocol clause is itself named that way. *Validity* denotes conceptual scientific properties of the benchmark; *adequacy* denotes the operational prospectively frozen gate used to decide whether protected primary execution was warranted.

Reader Map

The conference-facing main manuscript follows a problem–method–result–meaning structure and deliberately omits most development chronology. Table 1 directs readers to the corresponding detailed evidence. This separation changes presentation only: the frozen scientific record, Pilot-B decision and post-closure interpretation boundary are unchanged.

2 Detailed Benchmark Construction

2.1 Criteria and Capability Pathways

The six alternatives in the main manuscript were evaluated on the ten technology-agnostic criteria in Table 2. The original lifecycle-burden criterion was a cost; all other criteria were benefits.

The pre-specified capability mask in Table 3 defines synthetic functional pathways, not empirical impact magnitudes. Direct (D), indirect (I) and zero (0) pathways set broad response-amplitude bands.

2.2 Gaussian-Copula Contexts and Opportunity Functions

For demand, incident, person-movement, environmental and readiness factors, generate $z_{0,s}, z_{1,s}, \dots, z_{5,s} \stackrel{iid}{\sim} \mathcal{N}(0, 1)$ and set

$$\tilde{h}_{k,s}^{(\rho)} = \sqrt{\rho} z_{0,s} + \sqrt{1-\rho} z_{k,s}, \quad h_{k,s}^{(\rho)} = \Phi(\tilde{h}_{k,s}^{(\rho)}). \quad (1)$$

The same base draws were reused across dependence levels, while estimation and external TEST streams used disjoint namespaces. The first seven criterion opportunities were

$$\begin{aligned} o_{s1} &= h_D, & o_{s2} &= 0.60h_D + 0.40h_I, & o_{s3} &= 0.60h_D + 0.40h_T, \\ o_{s4} &= 0.50h_D + 0.50h_I, & o_{s5} &= h_I, & o_{s6} &= 0.60h_D + 0.40h_E, \\ o_{s7} &= 0.40h_D + 0.60h_T. \end{aligned} \quad (2)$$

Table 1. Navigation from the concise main argument to the detailed reproducibility record, with the evidential status of each block. Status codes are: EDA, executed development audit; PBA, Pilot-B confirmatory adequacy result; PCD, post-closure descriptive diagnostic; ANE, archived and not executed.

Main-paper element	Detailed supplementary evidence	Status
Controlled ITS laboratory	Criterion definitions, capability pathways, copula contexts and opportunity functions in Sect. 2.	EDA
Oracle and global attribution	Closed-form functional, joint-background moments, XGBoost, TreeSHAP and B100 checks in Sect. 3.	EDA
Structural correction	Calibration, positive control, numerical null and candidate screening in Sect. 4.	EDA
D29 decision geometry	World-resolved oracle orderings, winners, modal shares and regret in Sect. 5.	EDA
Pilot-B evidence (RQ2)	Per-world method contrasts, gate decision and objective-weight vectors in Sects. 5 and 7.	PBA
Observed selection maps (DQ1)	All audited modal identities, shares, unique-choice counts and pairwise agreement summaries in Sect. 5.	PCD
Decision operators and gate	Exact weighting, tie, regret and hard-stop rules in Sects. 6–7.	EDA
Unexecuted primary plan	Clearly labelled archival protocols in Sects. 8–12.	ANE
Integrity and closure	Content digests, software tests and permanent-stop record in Sects. 13–14.	EDA

Each replication seed r defined one benchmark world W_r : technology parameters, a master context stream and all common-random-number streams. The conditional method target had been

$$\Psi_m(x) = \mathbb{E}_W[\mathbb{E}_{H,\varepsilon}\{L_m(x; W, H, \varepsilon) \mid W\}]. \quad (3)$$

The world, rather than an alternative-context row, was the intended statistical unit. The primary worlds needed to estimate this quantity were never accessed after the gate fired.

2.3 Historical and Frozen Response Layers

For C_1 – C_7 , capability amplitudes were drawn once per world:

$$\theta_{aj} \sim \begin{cases} 0, & M_{aj} = 0, \\ \mathcal{U}(0.05, 0.20), & M_{aj} = I, \\ \mathcal{U}(0.20, 0.40), & M_{aj} = D. \end{cases} \quad (4)$$

The historical v2.1 response was

$$x_{asj} = \text{clip}(\theta_{aj} o_{sj} + \sigma_x \xi_{asj}, 0, 1). \quad (5)$$

Table 2. Controlled evaluation criteria.

Code	Criterion	Original direction
C_1	Mobility-efficiency improvement	Benefit
C_2	Travel-time reliability improvement	Benefit
C_3	Person-throughput improvement	Benefit
C_4	Safety-surrogate improvement	Benefit
C_5	Non-recurring-congestion resilience	Benefit
C_6	Environmental-efficiency improvement	Benefit
C_7	User and accessibility benefit	Benefit
C_8	Infrastructure and data readiness	Benefit
C_9	Interoperability and scalability	Benefit
C_{10}	Annualised lifecycle deployment burden	Cost

Table 3. Pre-specified structural capability mask.

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	Ready	Interop.	Burden
ATSC	D	D	D	I	I	I	I	M	H	M
TSP	I	D	D	I	0	I	D	M	H	L-M
TIDM	I	D	I	D	D	I	I	M	H	M
RMTI-DRG	I	I	I	I	D	I	D	L-M	H	L-M
V2X-CS	I	I	0	D	I	I	I	H	H	H
DSM	D	D	I	D	D	I	I	M-H	H	M-H

The deployment classes were

$$L \sim U(0.25, 0.40), \quad M \sim U(0.45, 0.60), \quad H \sim U(0.65, 0.80). \quad (6)$$

Compound classes were equal-probability mixtures. With readiness requirement r_a , interoperability ι_a and burden κ_a ,

$$\begin{aligned} x_{as8} &= \text{clip}(1 - \max(0, r_a - h_{R,s}) + \eta_{as8}, 0, 1), \\ x_{as9} &= \text{clip}(\iota_a + 0.10h_{R,s} + \eta_{as9}, 0, 1), \\ x_{as,10} &= \text{clip}(\kappa_a + 0.20(1 - h_{R,s})r_a + \eta_{as,10}, 0, 1). \end{aligned} \quad (7)$$

Benefit orientation used $g_{asj} = x_{asj}$ for $j \leq 9$ and $g_{as,10} = 1 - x_{as,10}$. At $\sigma_x = 0$, the first seven criteria reduced to $g_{asj} = \theta_{aj}o_{sj}$, which explains the exact normalization cancellation in the main paper.

The frozen D29 kernel was

$$g_{asj} = \theta_{aj}o_{sj} + 0.05v_{aj}o_{sj}(1 - o_{sj}), \quad j \leq 7. \quad (8)$$

For m_j active alternatives, its ordered contrast grid was $v_{\ell_j}^{grid} = -1 + 2(\ell - 1)/(m_j - 1)$. After sorting active IDs, one deterministic NumPy permutation used

$$\text{SeedSequence}([r, 2010, j]). \quad (9)$$

No outcome or winner information entered this assignment. Active $\theta_{aj} \geq 0.05$ ensured $\partial g / \partial o = \theta_{aj} + 0.05v_{aj}(1 - 2o) \geq 0$. Structural-zero pathways remained zero, and C_8 - C_{10} retained Eq. (7).

For $o_{sj} > 0$, within-context vector normalization over active alternatives gives the exact geometry

$$r_{asj} = \frac{\theta_{aj} + 0.05v_{aj}(1 - o_{sj})}{\sqrt{\sum_{b \in \mathcal{A}_j} [\theta_{bj} + 0.05v_{bj}(1 - o_{sj})]^2}}. \quad (10)$$

Thus D29 removes the v2.1 cancellation, but each criterion-specific normalized profile remains on a one-parameter curve indexed by o_{sj} . Because opportunities differ across criteria, Eq. (10) alone does not prove that a fixed-weight MOORA winner is context invariant. At Pilot-B closure, fixed-weight invariance was therefore an untested hypothesis rather than an explanation of the stop. The subsequently frozen descriptive selection-map diagnostic reported below resolves this question for the seven observed vectors only; it does not establish a causal effect of Eq. (10) or characterize unobserved weight vectors.

3 Oracle, Learning and Background Details

3.1 Oracle Utility and Closed-Form Attribution

The monotone transform, coefficient vector and interaction graph were

$$\begin{aligned} q_j(g) &= \frac{\log(1 + 2g)}{\log 3}, \\ \alpha &= (0.11, 0.04, 0.05, 0.07, 0.10, 0.14, 0.12, 0.16, 0.08, 0.13), \\ \mathcal{E} &= \{(1, 2), (3, 7), (4, 5), (6, 10), (8, 9)\}, \quad \beta_{jk} = 0.20. \end{aligned} \quad (11)$$

For $z_j = q_j(g_j)$, the empirical FIT-background moments were $\mu_j = \mathbb{E}_{bg}(z_j)$ and $\mu_{jk} = \mathbb{E}_{bg}(z_j z_k)$; dependence was retained by not replacing μ_{jk} with $\mu_j \mu_k$. The main-effect contribution was $\phi_j^{main} = \alpha_j(z_j - \mu_j)$. For pair $\beta_{jk} z_j z_k$, feature j received

$$\phi_j^{(jk)} = \frac{\beta_{jk}}{2} \{z_j \mu_k - \mu_{jk} + z_j z_k - \mu_j z_k\}, \quad (12)$$

with the symmetric expression for k . The closed-form attribution was

$$\phi_j^*(\mathbf{g}) = \frac{1}{1 + \lambda} \left[\alpha_j(z_j - \mu_j) + \lambda \sum_{k: (j, k) \in \mathcal{E}} \phi_j^{(jk)} \right], \quad (13)$$

with symmetric handling when j was second in a pair. Exhaustive coalition enumeration was a unit-test reference. Global weights normalized mean absolute WEIGHT-row attributions; total importance at most 10^{-12} made the vector undefined, with no Equal substitution.

3.2 XGBoost and TreeSHAP Freeze

XGBoost was selected once by context-grouped cross-validation on development FIT rows at the reference condition. Candidate 45 attained mean RMSE 0.017757058913365983 with

$$\begin{aligned} n_{\text{estimators}} &= 600, \quad d_{\text{max}} = 2, \quad \eta = 0.05, \\ \text{subsample} &= 0.8, \quad \text{colsample_bytree} = 0.8, \quad \text{min_child_weight} = 5, \\ \text{reg_alpha} &= 0, \quad \text{reg_lambda} = 1. \end{aligned} \quad (14)$$

The objective was `reg:squarederror`, with one computational thread and estimator random-state namespace 82002. The selected model was then fixed. Interventional TreeSHAP used 100 deterministic FIT identities selected with row-priority namespace 83001 and evaluated mean absolute attributions on WEIGHT. External TEST did not enter calibration, background selection or weighting.

3.3 Matched B100 Oracle Diagnostic

The primary oracle used all eligible FIT rows for μ_j and μ_{jk} . The secondary B100 diagnostic changed only these moments to the exact 100 frozen TreeSHAP identities; WEIGHT rows, the oracle functional and every other setting were unchanged. It compared full-FIT and B100 weight vectors through mean absolute difference, total variation, maximum absolute difference, Spearman correlation, Top-3/Top-5 overlap and maximum marginal/joint moment changes. Undefined vectors remained undefined. The only authorized pre-primary execution was seed 21001 at $N = 250$, $\rho = 0.4$ and $\lambda = 0.5$; no Y , SHAP package call or MCDM operation was involved.

4 Detailed Structural Calibration and Candidate Screening

4.1 Structural and Reachability Metrics

Let $G_j \in \mathbb{R}^{S \times A_j}$ denote the active-alternative response matrix for criterion j at zero measurement noise. If its singular values are $\sigma_{1j} \geq \sigma_{2j} \geq \dots$ and $E_j = \sum_{\ell} \sigma_{\ell j}^2$, the rank-one residual was

$$NS_j = 1 - \frac{\sigma_{1j}^2}{E_j}, \quad E_j > 10^{-24}. \quad (15)$$

No epsilon was inserted into this ratio; structural-zero matrices were reported as undefined. For each jointly positive active alternative pair,

$$LRV_{ab,j} = \text{SD}_s[\log(g_{asj}/g_{bsj})], \quad LRV50_j = \text{median}_{a < b} LRV_{ab,j}, \quad (16)$$

Table 4. Frozen D2.7 criterion-level scientific references.

Criterion	$T_{LRV,j}^{sci}$	$T_{NSV,j}^{sci}$
C_1	0.185606	0.101685
C_2	0.107107	0.069436
C_3	0.163029	0.064900
C_4	0.115796	0.069423
C_5	0.127964	0.073681
C_6	0.194899	0.129935
C_7	0.144080	0.073092

using population SD. For normalization family m ,

$$NSV_{j,m} = \left\{ S^{-1} \sum_s \left\| \mathbf{r}_{sj}^{(m)} - \bar{\mathbf{r}}_j^{(m)} \right\|_2^2 \right\}^{1/2}. \quad (17)$$

Vector normalization was the primary structural gate; sum, max and min-max variants were robustness diagnostics.

Reachability reflected one idiosyncratic Gaussian-copula shock while holding the shared and all other idiosyncratic shocks fixed. With opportunity functions recomputed after reflection,

$$RE_{kj} = \text{median}_{s,a \in A_j} |g_{asj}^{(-k)} - g_{asj}|, \quad SRE_{kj} = \frac{RE_{kj}}{\text{IQR}(g_j) + 10^{-12}}. \quad (18)$$

4.2 Scientific Positive Control

For active C_1 – C_7 pathways, the dedicated positive control was

$$g_{asj}^{PC,\ell} = \left[\theta_{aj} + 0.05c_a^{(\ell)}(2o_{sj} - 1) \right] o_{sj}, \quad (19)$$

with contrast base $(-1, -0.6, -0.2, 0.2, 0.6, 1)$ and all six cyclic rotations. Structural-zero pathways remained zero and C_8 – C_{10} were unchanged. Rotation medians were formed within development worlds 21001–21005 and world medians were then used as scientific references.

The paired positive-control uplift in mean normalized constant-modal regret was $(0.001485, 0.020022, -0.013836, 0.015534, 0.006161)$ for worlds 21001–21005. Because the activation rule required positive uplift in all five worlds, no Layer-B scientific regret reference was defined and no post-hoc value was introduced.

4.3 Numerical Null and Candidate Outcomes

Numerical thresholds were calibrated from 2000 exact rank-one replicates for each of five- and six-alternative active sets:

$$T_{NS}^{num} = 10^{-12}, \quad T_{LRV}^{num} = 10^{-10}, \quad T_{NSV}^{num} = 3.2506084454 \times 10^{-8}. \quad (20)$$

Table 5. Frozen D2.7 latent-pathway reachability references.

Pathway	$T_{SRE,kj}^{sci}$
$h_D \rightarrow C_1$	0.556245
$h_D \rightarrow C_2$	0.334437
$h_D \rightarrow C_3$	0.284878
$h_D \rightarrow C_4$	0.292844
$h_D \rightarrow C_6$	0.431612
$h_D \rightarrow C_7$	0.252576
$h_E \rightarrow C_6$	0.296869
$h_I \rightarrow C_2$	0.212043
$h_I \rightarrow C_4$	0.294905
$h_I \rightarrow C_5$	0.455271
$h_T \rightarrow C_3$	0.186995
$h_T \rightarrow C_7$	0.373380

All 35 historical v2.1 world-criterion rows fell below every threshold. For active pathways $j \leq 7$, the D4-F1 curvature family was $g_{asj}^{D4F1}(\chi) = \theta_{aj} o_{sj} [1 + \chi v_{aj} (1 - o_{sj})]$, with structural-zero pathways fixed at zero. It cleared some structural references at higher curvature but lost reachability; no $\chi \in \{0.1, \dots, 1.0\}$ passed all gates, so the family was rejected.

The later exact D2.9 audit evaluated all $6^5 = 7776$ prespecified positive-control profile assignments. None met the old simultaneous binary conjunction. This established non-viability for that calibrated design, not mathematical impossibility. Under the corrected prospective rule, 54 candidates were eligible. D29 was retained solely for protocol continuity and minimal intervention, not scientific preference, downstream performance, SHAP pattern or ITS identity.

The count of 54 is composed rather than directly observed in a single artefact, and is decomposed here to prevent confusion. The nonselective retrospective re-adjudication examined 54 historical parameter points and returned 44 VERIFIED_ELIGIBLE, 0 VERIFIED_INELIGIBLE and 10 INSUFFICIENT_FROZEN_EVIDENCE. The 10 unresolved points were all v2.2 points whose frozen artefact schema did not record C8–C10 invariance. A separate supplemental resolution then established that invariance from the contemporaneous frozen source contract, and all 10 satisfied the corrected hard constraints. The frozen retention artefact therefore records an eligible universe of $44 + 10 = 54$ specifications. Eligibility is explicitly non-unique: the retention of D29 is a minimal-intervention and provenance-stability decision, not a superiority, uniqueness or statistical-selection claim, and no ranking or re-optimization over the 54 points was performed.

Table 6. Complete frozen D29 Layer-B development record.

World	Orderings	Winners	Modal	Modal share	Mean regret	Median regret	P_{95} regret
21001	23	2 A_6		0.983	0.00042018	0	0
21002	78	5 A_1		0.519	0.05992337	0	0.29155475
21003	24	4 A_2		0.745	0.04523642	0	0.28747533
21004	17	3 A_2		0.827	0.01734351	0	0.12641643
21005	38	4 A_2		0.809	0.01606539	0	0.12654230

5 Detailed Development Audit Results

5.1 D29 Layer-B Decision Geometry

The one-shot characterization used development worlds 21001–21005, the complete $N = 1000$ FIT+WEIGHT pool, $\rho = 0.4$, $\sigma_x = 0$ and $\lambda = 0.5$. It generated no Y and executed no learned model, SHAP package or MCDM method. Regret used the exact context-specific oracle range and was undefined only if that range was at most 10^{-12} ; no context was undefined.

Across worlds, the medians were 24 orderings, four winners, modal share 0.809, mean regret 0.01734351, median regret zero and P_{95} regret 0.12654230. At the directly comparable world 21001, the historical v2.1 record had 19 orderings, two winners, modal share 0.962 and mean always-modal regret 0.0011106 under its frozen $D_s + 10^{-12}$ denominator. D29 had 23 orderings, two winners, modal share 0.983 and mean regret 0.00042018 under exact D_s . The comparison therefore supports neither universal degeneration nor universal improvement.

5.2 XGBoost and TreeSHAP Development Checks

The selected XGBoost hyperparameters are reported in Eq. (14); candidate number and grouped-CV RMSE are given above. At world 21001 and the reference condition, total mean-absolute TreeSHAP importance was 0.07006131011183041 and maximum local-accuracy error was 6.45×10^{-7} . The development-only weight vector for $(C_1, \dots, C_{10})$ was

$$(0.124860, 0.113194, 0.069426, 0.025945, 0.155169, \\ 0.078339, 0.071621, 0.265059, 0.020729, 0.075658). \quad (21)$$

These observations did not tune comparator definitions, MCDM rules or winner semantics.

5.3 B100-Matched Background Result

Both oracle vectors were defined, and the 100 identities exactly matched the frozen TreeSHAP membership. Total mean-absolute importance was 0.06961866590 under full-FIT moments and 0.06992189186 under B100 moments.

Table 7. Development-only oracle-attribution weights under full-FIT and matched B100 moments.

Criterion	Full-FIT	B100	Absolute difference
C_1	0.09663438	0.09712037	0.00048600
C_2	0.04849942	0.04854063	0.00004122
C_3	0.05910838	0.05913637	0.00002799
C_4	0.06665455	0.06559847	0.00105608
C_5	0.14880967	0.14667297	0.00213669
C_6	0.09368358	0.09388953	0.00020596
C_7	0.07615062	0.07594850	0.00020212
C_8	0.24766693	0.25232065	0.00465372
C_9	0.04819959	0.04935317	0.00115358
C_{10}	0.11459290	0.11141932	0.00317357

Table 8. Per-world closed-form oracle-attribution and SHAP advantages over MajorityWinner.

World	Oracle Δ Top-1	SHAP Δ Top-1	Oracle Δ regret	SHAP Δ regret
21001	0.000	0.000	0.000000	0.000000
21002	0.035	0.030	0.001538	0.003064
21003	-0.465	0.020	-0.143522	0.006717
21004	-0.010	0.000	-0.002119	0.000000
21005	-0.380	-0.035	-0.090328	-0.002986

The weight-vector discrepancies were

$$\text{MAE}_w = 0.0013136928, \quad TV_w = 0.0065684640, \quad \max_j |\Delta w_j| = 0.0046537223. \quad (22)$$

Spearman correlation was 0.98787879, and Top-3 and Top-5 overlaps were both 1.0. Maximum absolute marginal- and joint-moment changes were 0.0098907938 and 0.0041811926, respectively. Equal fallback was not used. These results show small finite-background sensitivity at this one development reference, not uniform robustness.

5.4 World-Resolved Pilot-B Oracle and SHAP Comparison

Table 8 retains the full comparison moved from the main paper. Each entry is an advantage over the world-specific MajorityWinner reference; positive values favour the structured method.

The closed-form method exactly implements the frozen global interventional attribution functional; it is not an exact representation of context-specific oracle utility. Thus its failure to escape MajorityWinner is evidence about the frozen global-compression-MOORA pipeline, not a numerical error in the attribution formula.

Table 9. Frozen objective-weight vectors for development world 21001.

Method	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10
CRITIC	.079	.075	.101	.093	.150	.099	.097	.111	.084	.111
Entropy	.131	.114	.175	.099	.236	.064	.084	.027	.032	.038

5.5 Illustrative Frozen Objective-Weight Vectors

Table 9 directly transcribes the CRITIC and Entropy vectors stored for development world 21001; no new transformation of TEST decisions was performed. The display cannot alter the frozen gate or establish a performance mechanism. Both vectors were defined without Equal fallback, and neither has its largest component on C8 or C10. Thus concentration on those two criteria is not supported in this world. The table is deliberately not combined with five-world median performance to infer cross-world geometry.

The aggregate coincidence in world 21003 did not, by itself, identify a common decision map. Equal, CRITIC and Entropy each recorded Top-1 accuracy 0.190 and mean normalized regret 0.316758, but their mean Kendall τ_b values were respectively 0.5493, 0.5427 and 0.5113. Top-1 accuracy and normalized regret depend on the selected alternative, whereas Kendall τ_b depends on the complete ranking. The τ_b differences therefore establish different lower-order ranking behaviour on average but are logically uninformative about argmax equality. That ambiguity motivated the separately frozen diagnostic below.

5.6 Frozen Post-Closure Fixed-Weight Selection-Map Diagnostic (DQ1)

Evidential status: post-closure, descriptive, non-confirmatory. This subsection answers DQ1 only. It cannot modify, reinterpret, rescue or reopen the Pilot-B gate, it defines no pass/fail criterion, and none of its content is confirmatory evidence for RQ1 or RQ2.

The diagnostic was specified only after the Pilot-B result had been independently audited, frozen and permanently closed. Its protocol was committed before any context-level fixed-weight result was reconstructed. It used worlds 21001–21005 at the same reference condition and the 200 frozen external TEST identities per world. The six structured vectors were read from the immutable Pilot-B result without re-estimation; Equal was exactly $(0.1, \dots, 0.1)$. D29 TEST decision matrices and oracle utilities were reconstructed only to validate the archived aggregates. No target Y , model fit/prediction, SHAP, PI, RandomWeights, bootstrap or protected seed was executed.

For each world, method and context the result stores the selected alternative; all six MOORA scores and ranks; oracle utilities, ranks and selected alternative; Top-1 match; normalized regret contribution; and agreement with the frozen MajorityWinner identity. Exactly $5 \times 7 \times 200 = 7000$ records were independently audited. The audit reconstructed every selection, anchor-based rank and regret

Table 10. Audited selection maps for the seven observed fixed vectors. Each method entry is modal alternative/modal share/number of distinct selected alternatives over 200 TEST contexts.

World	MW	Oracle	SHAP	PI	Ridge+	CRITIC	Entropy	Equal
21001	A6	A6/1.000/1	A6/1.000/1	A6/.980/2	A6/1.000/1	A6/1.000/1	A6/1.000/1	A6/1.000/1
21002	A1	A1/.700/3	A1/.950/2	A1/.630/2	A1/.875/3	A3/.800/3	A1/1.000/1	A1/1.000/1
21003	A2	A6/.720/4	A2/.690/3	A2/.530/3	A6/.495/3	A6/1.000/1	A6/1.000/1	A6/1.000/1
21004	A2	A2/.990/2	A2/1.000/1	A2/.775/4	A2/.990/2	A3/.750/3	A3/.985/3	A2/1.000/1
21005	A2	A6/.350/4	A2/.790/3	A3/.675/5	A6/.375/4	A6/.915/3	A6/.800/3	A2/1.000/1

directly from the stored full-precision vectors, matched all 35 archived method-world aggregates within 10^{-12} , and rebuilt all 105 pairwise selection-agreement summaries without executing the generator.

Table 10 reports each method’s modal selected alternative, modal share and number of distinct selected alternatives. Entries use the format “alternative/share/ k ”, where $k = 1$ denotes context-invariant selection in that world. MajorityWinner (MW) is a single frozen alternative by construction.

Equal was context invariant in all five worlds and exactly matched MajorityWinner across all 200 contexts in worlds 21001, 21002, 21004 and 21005. In world 21003, Equal instead selected A6 throughout while MajorityWinner was A2. The numbers of constant-selection worlds were Oracle 1, SHAP 2, PI 0, Ridge+ 1, CRITIC 2, Entropy 3 and Equal 5; exact all-context MajorityWinner agreement occurred in 1, 2, 0, 1, 1, 2 and 4 worlds, respectively.

The world-21003 ambiguity is now directly resolved: Equal, CRITIC and Entropy all selected A6 in every TEST context. Their different mean τ_b values therefore reflect differences below the argmax, not different selected alternatives. More broadly, the heterogeneous constancy counts show that fixed-vector MOORA was neither universally context invariant nor uniformly context responsive over the seven observed vectors. The result supports coarse observed-vector decision regions as a descriptive hypothesis, but cannot establish the causal mechanism of the Pilot-B stop, characterize unobserved regions of the weight simplex, rank methods or generalize beyond these five development worlds.

The new artefact makes paired context resampling technically possible, but the prospectively frozen diagnostic explicitly excluded bootstrap and defined no inferential or pass/fail role. No resampling was executed. Any future paired sensitivity analysis requires its own protocol, frozen before inspecting bootstrap output, and cannot revise the closed Pilot-B decision.

6 Detailed Weighting and Decision Operators

The primary information partitions were fixed: Oracle, SHAP, PI, CRITIC and Entropy used WEIGHT; Ridge+ used FIT; no method used external TEST for fitting or weight estimation. Oracle and SHAP normalized mean absolute attributions. SHAP was declared undefined if total importance was at most 10^{-12} and was never replaced by Equal weights.

CRITIC applied criterion-wise min–max scaling, population SD and Pearson conflict over nonconstant columns. Entropy applied the same min–max scaling followed by Shannon diversification with $0 \log 0 = 0$. Ridge+ standardized predictors and target with fold-local population moments, solved non-negative L2-augmented least squares without an intercept, and used the frozen grid

$$\tau \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}. \quad (23)$$

Five-fold context-grouped original-scale RMSE selected τ , with the smallest value as tie break.

Permutation importance kept the fitted XGBoost model fixed. For each criterion, it copied the complete six-alternative block from source context $\pi(d)$ to destination context d under 20 unique derangements. Their frozen generator was

$$\text{SeedSequence}([\text{replication}, 84001, N]). \quad (24)$$

The same derangements were reused across criteria and factor levels for a given seed and N . Importance was permuted minus baseline WEIGHT MSE; negative means were truncated before normalization and repeat dispersion used sample SD. CRITIC, Entropy, Ridge+ and PI used Equal weights only under their frozen zero-information rules, with fallback recorded.

Equal was $(0.1, \dots, 0.1)$. RandomWeights used 200 Dirichlet $(1, \dots, 1)$ vectors per world from namespace 81001 and reused them across conditions. MajorityWinner estimated the oracle modal winner on WEIGHT and applied that constant choice to TEST. MajorityWinner was intentionally oracle-informed: it was the context-free oracle-modal alternative estimated on WEIGHT and evaluated on TEST, and it was optimal only for empirical oracle Top-1 frequency among constant choices. It is not claimed to minimize regret or to constitute an operational transportation policy. It was neither a deployable competitor nor a universally optimal choice, and it was not stakeholder-derived or normative. The regret arm of the frozen gate uses this same prospectively frozen Top-1-derived constant reference rather than a separately optimized regret-minimizing constant policy; a future benchmark could prospectively define metric-matched constant references. The frozen gate itself is unchanged by this observation. Direct-XGBoost was a parallel predictive-decision reference rather than a weighting method. MOORA used within-context vector normalization and a weighted sum; TOPSIS used the same normalized benefit matrix as a conditional aggregation robustness operator.

6.1 Frozen Tie Semantics and Archived Decision Margin

Within each context, alternatives were sorted by descending raw score and ascending `alternative_id`. The highest unassigned score anchored a tie group; every remaining alternative within absolute distance 10^{-12} of that anchor joined the group and received the average of the occupied one-based rank positions. Ties were not chained through adjacent near-equalities. Deterministic Top-1 used

$$\mathcal{T}_s = \{a : S_{\max, s} - S_{as} \leq 10^{-12}\} \quad (25)$$

Table 11. Complete frozen Pilot-B gate decision.

Gate	Audited result	Stop
Coverage	Six structured methods eligible without fallback	No
Random equivalence	Four of six methods separated from RandomWeights	No
Modal-winner effective tie	Zero of six methods escaped MajorityWinner	Yes
Oracle-winner dominance	Pooled modal share $0.526 < 0.90$	No
Weight insensitivity	Invariant fraction $0.004 < 0.90$; width-qualified worlds $0 < 4$	No
Overall OR rule	At least one frozen stop fired	Yes

and selected the smallest alternative ID in $\mathcal{T}_s$. Relative tolerance was zero. These semantics applied to oracle, MOORA, TOPSIS and Direct-XGBoost scores.

The conditional primary plan also defined the guarded oracle decision margin

$$M_s = \frac{U_{(1),s}^* - U_{(2),s}^*}{\max_a U_{as}^* - \min_a U_{as}^* + 10^{-12}}. \quad (26)$$

Margin-conditioned analyses were not executed because Pilot B stopped the primary study.

7 Frozen Pilot-B Gate and Complete Decision

Pilot B reused development worlds 21001–21005 at the frozen reference

$$(N, c, \rho, \lambda) = (250, 0.30, 0.4, 0.5). \quad (27)$$

Each world used 200 external TEST contexts and 200 common RandomWeights draws per world. It was a protocol-level adequacy screen, not independent population validation. A structured method was eligible only if it was defined without Equal fallback in all five worlds and Kendall’s τ_b was defined in at least 190 TEST contexts per world; at least 190 RandomWeights draws also had to be eligible. Cross-world advantages used medians and required strict positivity in at least four worlds.

The five independently sufficient hard stops were: incomplete coverage; no eligible method clearing median $\Delta\tau_b = 0.05$ or $\Delta\text{mean-regret} = 0.01$ versus RandomWeights; no eligible method clearing median $\Delta\text{Top-1} = 0.02$ or $\Delta\text{mean-regret} = 0.01$ versus MajorityWinner; pooled oracle modal share at least 0.90; or the frozen random-weight insensitivity criterion, including its 0.90 invariant fraction and joint width limits 0.05 (τ_b), 0.02 (Top-1) and 0.01 (mean regret) in at least four worlds. Passage required every stop to be false.

The best strictly positive consistency count was two of five worlds for either Top-1 or mean regret, and the best cross-world medians were zero. The independently reconstructed decision therefore barred the protected factorial. This is a negative result about the adequacy of the frozen comparison at one condition, not evidence that SHAP, weights or MCDM fail universally.

Table 12. Archived nested estimation partitions and fixed TEST size.

N FIT WEIGHT external TEST			
25	20	5	200
50	40	10	200
100	80	20	200
250	200	50	200
1000	800	200	200

8 Archived Stage-Wise Fidelity Ladder

The conditional primary analysis contained nine representations:

1. OracleUtility, the full interacting oracle;
2. OracleMainEffectReference, with the interaction removed;
3. OracleGlobalAttributionNonlinearQ;
4. SHAPGlobalAttributionNonlinearQ;
5. OracleGlobalAttributionLinearG;
6. SHAPGlobalAttributionLinearG;
7. OracleWeightMOORA;
8. SHAPWeightMOORA;
9. DirectXGBoost, a parallel reference rather than a sequential stage.

Eight paired contrasts isolated interaction removal, oracle global compression, SHAP estimation on the nonlinear scale, nonlinear-to-linear transformation under each weight vector, MOORA transformation under each vector and post-MOORA Oracle-versus-SHAP attribution. Kendall’s τ_b and normalized oracle regret were specified for every representation. The ladder was diagnostic, not an additive error decomposition, and was not executed because Pilot B stopped the primary study.

9 Conditional Primary Factorial

The four planned primary factors were

$$N \in \{25, 50, 100, 250, 1000\}, \quad c \in \{0.10, 0.30, 0.60\}, \quad (28)$$

$$\rho \in \{0, 0.4, 0.8\}, \quad \lambda \in \{0, 0.5, 1\}. \quad (29)$$

The resulting $5 \times 3 \times 3 \times 3 = 135$ conditions were to be repeated over 30 prespecified primary benchmark-instance seeds, giving 4050 seed-condition replications. The reference condition was

$$(N, c, \rho, \lambda) = (250, 0.30, 0.4, 0.5). \quad (30)$$

Technology-response parameters and common-random-number streams were to be sampled once per seed and reused across factor levels. For each (ρ, λ) cell,

Table 13. Frozen 200-world mean attribution-scale dispersion.

Criterion	Mean dispersion
C_8	0.122873493
C_{10}	0.119403289
C_5	0.110114266
C_2	0.094820283
C_3	0.090482741
C_4	0.088452971
C_1	0.087135074
C_7	0.078706704
C_6	0.049591336
C_9	0.028266887

the oracle signal scale s_U was to be calculated on the complete 1000-context FIT+WEIGHT master pool so that changing N would not change the imposed noise-to-signal ratio. Realized signal-to-noise ratio was to be logged for every cell. FIT, WEIGHT and external TEST partitions were to remain context-disjoint, and no TEST row was to be used for model fitting or weight estimation.

10 Archived Alpha-Dispersion Stress Analysis

The conditional primary analysis retained the frozen heterogeneous coefficient vector. At the reference condition only, three additional coefficient structures had been fixed: balanced, dispersion-aligned and dispersion-anti-aligned. The mapping was obtained from an independent 200-world technology ensemble using

$$D_{j,w}^{attr} = \frac{1}{M_w} \sum_i |q_j(g_{ijw}) - \bar{q}_{j,w}|, \quad \bar{D}_j^{attr} = \mathbb{E}_W[D_{j,W}^{attr}]. \quad (31)$$

The complete ordering agreed between 150 and 200 worlds:

$$C_8 > C_{10} > C_5 > C_2 > C_3 > C_4 > C_1 > C_7 > C_6 > C_9. \quad (32)$$

Balanced coefficients were $\alpha_j = 0.10$. The aligned and anti-aligned mappings were

$$\begin{aligned} \boldsymbol{\alpha}^{aligned} : & (C_8, C_{10}, C_5, C_2, C_3, C_4, C_1, C_7, C_6, C_9) \\ & \mapsto (0.16, 0.14, 0.13, 0.12, 0.11, 0.10, 0.08, 0.07, 0.05, 0.04), \end{aligned} \quad (33)$$

$$\begin{aligned} \boldsymbol{\alpha}^{anti} : & (C_9, C_6, C_7, C_1, C_4, C_3, C_2, C_5, C_{10}, C_8) \\ & \mapsto (0.16, 0.14, 0.13, 0.12, 0.11, 0.10, 0.08, 0.07, 0.05, 0.04). \end{aligned} \quad (34)$$

The frozen primary heterogeneous assignment had weak rank association with the dispersion ordering ($\rho_S = 0.1273$, $\tau_b = 0.0222$), so it was not changed. Although the D3 mapping itself was a completed development artefact, the planned primary stress comparisons were never executed.

11 Archived Bootstrap and Robustness Plan

At the reference condition, $B = 200$ context-level bootstrap replications were to resample FIT and WEIGHT contexts separately, refit the model, recompute attributions and weights, and evaluate decisions on the fixed original external TEST contexts. Confidence intervals were to be reported for criterion weights, total-variation weight error, Kendall's τ_b , Top-1 accuracy and normalized oracle regret. No cross-context probability that a particular ITS was ranked first was to be interpreted as a global deployment probability.

A 50-oracle ensemble was to leave the frozen D29 technology-response kernel unchanged while varying oracle coefficient vectors, nonlinearities and semantically admissible interaction pairs. Relative outcome noise was to be recalibrated to each oracle variant's signal standard deviation. This ensemble would not replace the response generator and therefore would not reintroduce the historical v2.1 separability defect through oracle-only perturbations.

Two distribution-shift analyses were specified: holding out the upper 20% of demand pressure and holding out the lower 20% of infrastructure readiness. Secondary robustness analyses were to include criterion removal, $\pm 10/20/30\%$ SHAP-weight perturbations, leave-one-alternative-out analysis and MOORA-TOPSIS comparison.

12 Archived Statistical Analysis Plan

Factor effects were to be summarized with repeated-seed regression or mixed-effects models, emphasizing effect sizes and uncertainty intervals rather than binary significance alone. Method comparisons were paired because every method would share the same benchmark instance. The 30 primary seeds were to be treated as benchmark-instance clusters; alternative-context rows were not to be interpreted as independent superpopulation replications.

Method availability was to be reported explicitly. For each factorial cell, undefined Oracle/SHAP weight rates and recorded Equal-fallback rates were to accompany conditional decision-fidelity summaries. Pairwise contrasts were to use only benchmark instances on which both methods were defined, with coverage and fallback rates reported alongside them. Undefined Oracle or SHAP vectors were never to be silently replaced by Equal weights.

Planned outputs included predictive-performance and Direct-XGBoost references, weight-recovery tables, decision-fidelity summaries, factorial effect summaries, alpha-alignment stress results, bootstrap intervals, validity-boundary figures and the prespecified stage-wise fidelity ladder. None of these primary outputs exists under v1.

13 Reproducibility and Content-Integrity Record

Table 14 gives non-identifying truncated content digests. The selected B100 background exactly matched the frozen TreeSHAP identity set.

Table 14. Truncated, non-identifying content digests of the frozen artefacts underlying the reported numbers.

Artifact	Truncated SHA-256
Oracle B100 result	7f9864bd...439c9fb0
TreeSHAP background identities	3184c675...1646eda1
Pilot-B result	05072857...9f787a88
Fixed-weight selection-map result	a1f293b6...d7e20f4e

The full anonymous provenance package records, in order, the threshold freeze, evaluator/executor freeze, scientifically invariant direct-file loader repair, audited Pilot-B result freeze, immutable closure reference, post-closure diagnostic protocol freeze, implementation freeze and independently audited result freeze. The worktree was clean after each closure point, and the final complete test run reported 836 passes with three pre-existing external plotting warnings. Exact repository identifiers are intentionally omitted here to protect anonymous review. These facts document software and artefact integrity; they are not statistical evidence.

14 Prospective Stop and Permanent Closure

Pilot B was governed by five independently sufficient hard stops: coverage failure, failure to separate RandomWeights, failure to escape MajorityWinner, pooled oracle-winner dominance and weight insensitivity. Passage required every flag to be false. In the frozen result, only the modal-winner effective-tie flag was true, but the overall OR rule therefore barred every analysis in this supplement.

The immutable version-control, digest and test record appears in Section 13. Pilot B v1 is closed and must not be rerun. Any later study requires a new specification and version reference, explicit disclosure of the v1 failure and development-only design work; it cannot alter the v1 thresholds, result or closure reference. The fixed-weight selection-map diagnostic in Section 5.6 is such a separately frozen descriptive layer; it neither reopened Pilot B nor authorized any protected analysis.